Expt No:
01
Date:
28 / 01 /2026

ALPHANUMERIC AND SYMBOLIC TRAJECTORY SYNTHESIS

1. **AIM:**

   To design and implement a high-precision robotic trajectory for the synthesis of alphanumeric or symbolic contours through the orchestration of linear and circular interpolation techniques on a 6-axis industrial manipulator.

2. **APPARATUS REQUIRED:**
   a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
   b) Control Hardware: CR800-D series dedicated robot controller.
   c) Software Suite: RT ToolBox3 integrated engineering environment.
   d) End-of-Arm Tooling : Precision drawing effector configured within the tool coordinate system.
   e) Infrastructure: Solid installation platform and standard TCP/IP or standard USB communication interface.

3. **THEORY:**

   Trajectory generation for planar symbolic mapping requires the mathematical decomposition of characters into primitive geometric segments.
   a) Core Principle:

      The robot Tool Center Point (TCP) is commanded to follow a continuous path synthesized from individual motion commands that dictate both spatial orientation and execution velocity.
   b) Working Mechanism:
      i) Linear Interpolation (Mvs): The system calculates a straight-line motion vector between two spatial coordinates, maintaining constant velocity. This is essential for rendering the rigid segments of characters (e.g., 'L', 'H').
      ii) Circular Interpolation (Mvc / Mvr): Utilizing a three-point reference logic (Start, Auxiliary/Passing, and End), the controller mathematically generates the rotational arcs required for curved characters (e.g., 'O', 'S').
      iii) Continuous Path Control (Cnt): By invoking Cnt 1, the controller bypasses deceleration at intermediate via-points. This ensures fluid motion blending across trajectory nodes, thereby preserving geometric integrity and minimizing cycle time.

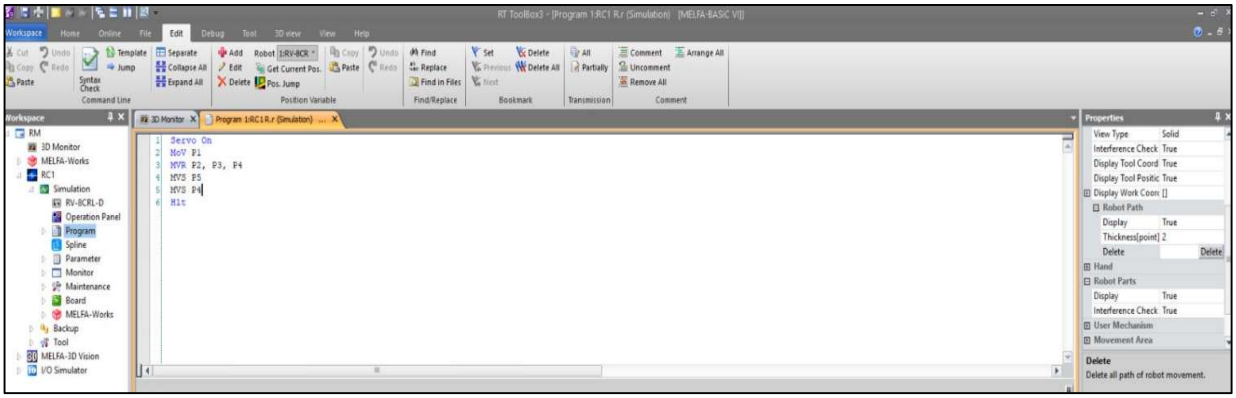

Figure 1: Source Code for Robotic Interpolation and Trajectory Planning (R Letter).

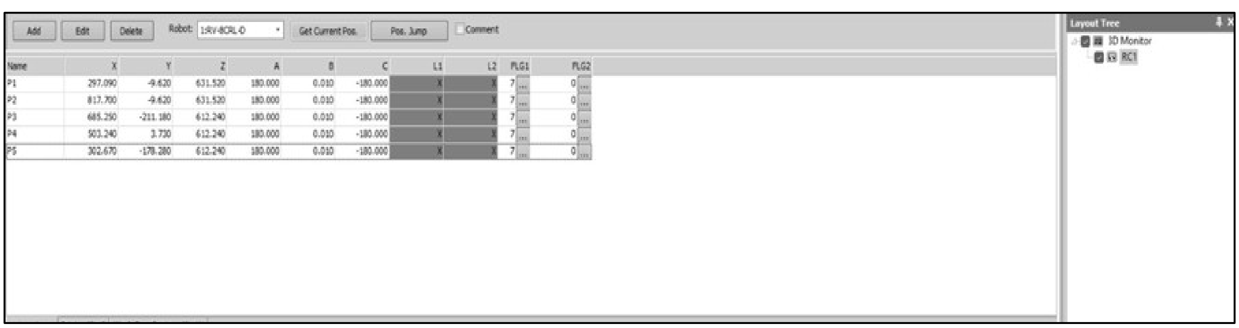

Figure 2: Defined Workspace Coordinates for Trajectory Synthesis.

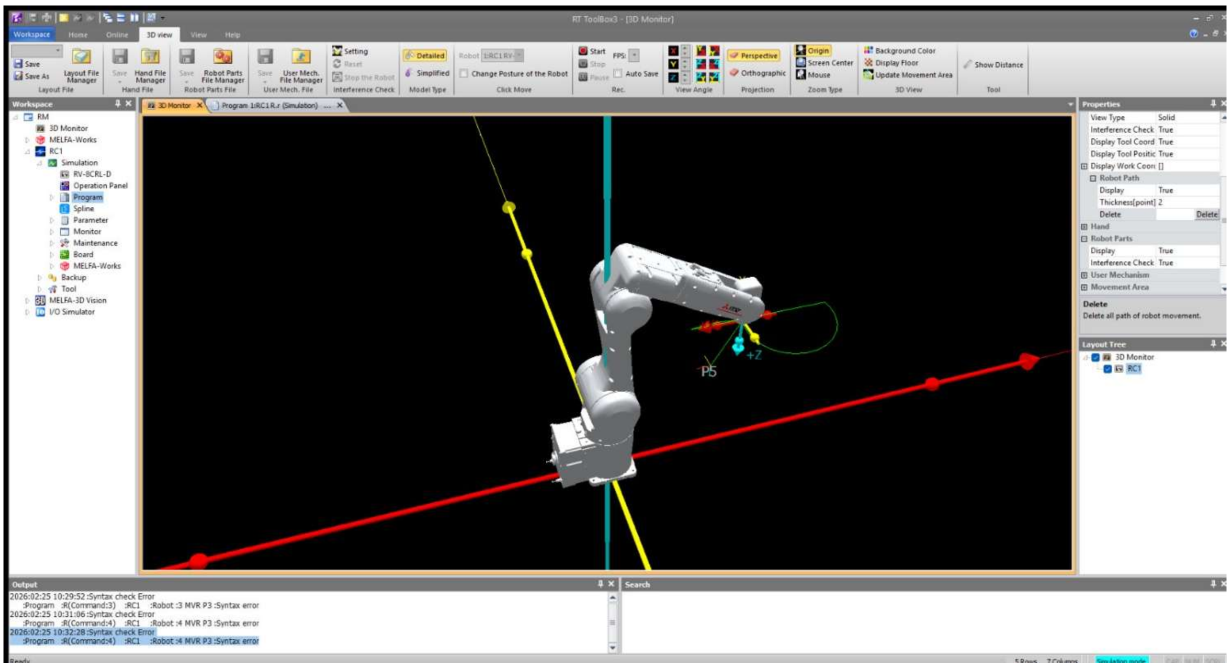

Figure 3: Simulated Kinematic Trajectory in the RT ToolBox3 3D Monitor**.**

4. PROCEDURE:

a) Setup & Connections:

Establish the communication link between the engineering workstation and the CR800-D controller. Perform an origin data verification to ensure absolute axis calibration.

b) Algorithm & Logic:

i) Initialize servos and designate the drawing tool parameters.
ii) Execute a joint-interpolated move to a stabilized approach height directly above the initial character coordinate.
iii) Sequentially trigger interpolation commands (Mvs for straight vectors, Mvc for curves) to construct the symbol.
iv) Incorporate temporal delays (Dly) between active strokes to allow for mechanical vibration settling.
v) Terminate the cycle by retreating to a verified safe home position (phome).

c) Execution Steps:

i) Load the synthesized logic into the RT ToolBox3 program editor.
ii) Perform a Step Run in manual mode at reduced override speed to validate the path and ensure collision avoidance.
iii) Transition the controller to Automatic mode for high-speed trajectory execution.

d) Observation Instructions:

Utilize the RT ToolBox3 Oscillograph tool to analyze motor current commands and axis load levels, specifically noting load spikes during high-speed curve transitions.

5. RESULT AND OBSERVATION:

The system achieved accurate contour synthesis with smooth motion and consistent precision throughout the operation, as verified by faculty evaluation.

Expt No:
02
Date:
04 / 02 /2026

KINEMATIC ALIGNMENT LOGIC AND MULTI-POINT SPATIAL MAPPING

1. AIM:

To design and implement kinematic alignment strategies for high-precision robotic assembly tasks (Peg-in-Hole) utilizing intermediate alignment waypoints and axial approach vectors on a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
b) Control Hardware: CR800-D series dedicated robot controller.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) End-of-Arm Tooling : Precision-machined assembly base with high-tolerance receptacle holes and a corresponding "peg" end-effector.
e) Infrastructure: Solid installation platform and standard TCP/IP or standard USB communication interface.

3. THEORY:

High-precision insertion tasks require the workpiece to be perfectly coaxial with its receptacle before final seating to prevent mechanical jamming or component damage.

a) Core Principle:

Kinematic alignment utilizes a sequence of spatial waypoints to transition the Tool Center Point (TCP) from a general workspace area to a specific assembly axis. This ensures that the final insertion vector is purely axial.

b) Working Mechanism:

i) Intermediate Waypoints (pint, pint1, pint2): These points serve as transition nodes that guide the robot to the insertion axis while clearing environmental obstacles.
ii) Axial Approach via Linear Interpolation (Mvs): Once alignment is achieved, depth-based moves must be executed using Mvs. This ensures the TCP moves in a rigid straight line, maintaining the vertical orientation required for high-tolerance seating.
iii) Coaxial Retraction: The exit path mirrors the insertion path linearly to avoid shearing forces against the receptacle walls.

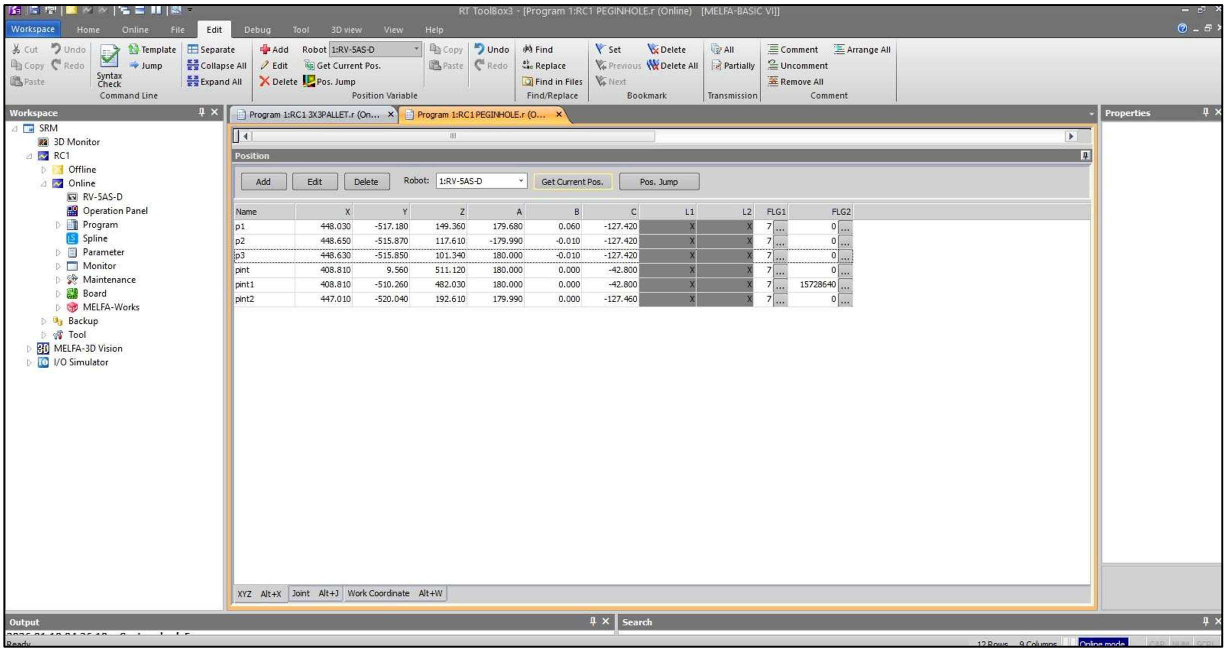

Figure 1: Position Variable table in RT ToolBox3 showing waypoint coordinates (pint series) and insertion points (p1–p3).

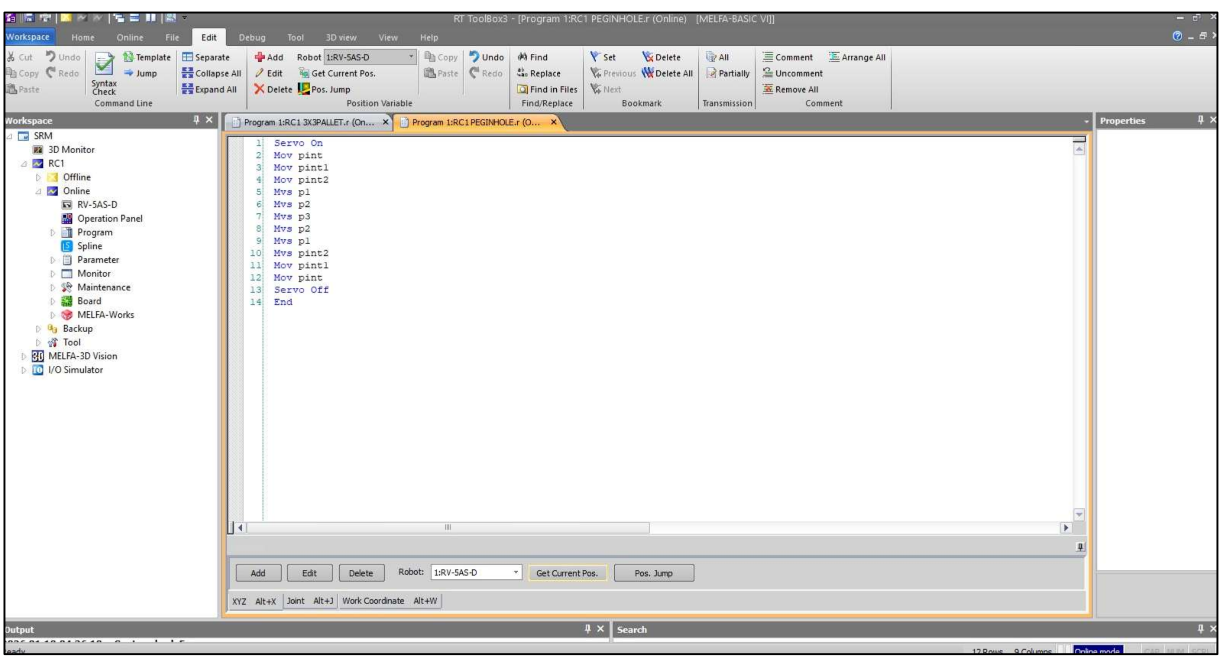

Figure 2: Program Editor in RT ToolBox3 with MELFA-BASIC VI code, highlighting the shift from joint (Mov) to linear (Mvs) interpolation.

Figure 3: Mitsubishi RV-5AS-D manipulator aligned above the fixture before insertion.

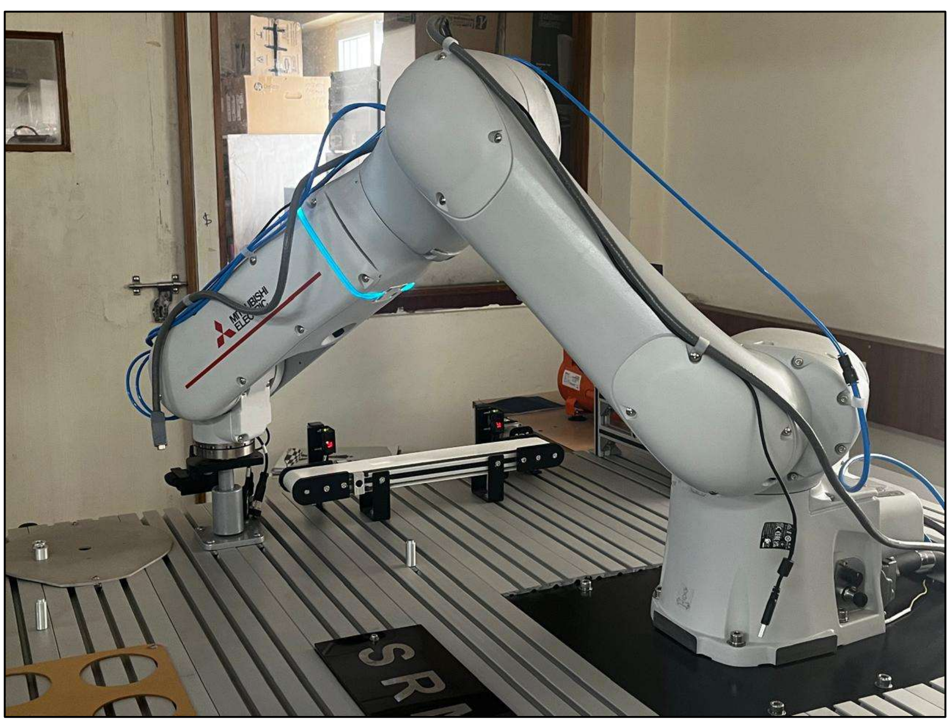

Figure 4: Live peg-in-hole assembly on the test bench.

4. PROCEDURE:

a) Setup & Connections:

Establish communication between RT ToolBox3 and the CR800 controller. Secure the assembly fixture and manually jog the robot using the teach pendant to define the seated assembly positions (p1, p2, p3) and the alignment nodes (pint, pint1, pint2).

b) Algorithm & Logic:

i) Enable system servos and move to the global intermediate waypoint (pint).

ii) Navigate through axis-alignment waypoints (pint1, pint2) to establish the approach vector.

iii) Execute high-precision linear insertion through seating depths (p1 to p3).

iv) Retract along the same axial path using linear interpolation.

v) Depart through the alignment waypoints to the home position and disable servos

c) Execution Steps:

i) Perform a Step Run in manual mode to verify that the axial alignment between pint2 and p1 is perfectly vertical.

ii) Check for potential "Axis Load" spikes using the Oscillograph tool during the linear depth moves (p1-p3) to ensure no mechanical binding occurs

iii) Transition to Automatic mode for operational repeatability testing.

d) Observation Instructions:

Visually monitor the insertion to ensure there is no lateral deflection of the manipulator arm. Verify the precision of the TCP relative to the target receptacle using the Cur pos display in the software.

5. RESULT AND OBSERVATION:

The robotic system successfully achieved high-precision kinematic alignment. The use of multi-point mapping and rigid linear interpolation ensured consistent coaxial insertion and retraction without mechanical interference, as verified by functional evaluation.

Expt No:
03
Date:
18 / 02 /2026

DYNAMIC PATH SYNTHESIS AND CONSTANT VELOCITY CONTROL

1. AIM:

To construct continuous, non-linear trajectories for industrial dispensing (glueing) applications and analyze the impact of constant velocity control and path blending on trajectory fidelity using a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
b) Control Hardware: CR800-D series dedicated robot controller.
c) Software Suite: RT ToolBox3 integrated engineering environment with Oscillograph tool enabled.
d) End-of-Arm Tooling : Simulated dispensing nozzle or glue gun configured within the tool coordinate system
e) Infrastructure: Practical board for glueing paths and standard communication interface.

3. THEORY:

Industrial applications such as glueing, sealing, and coating require a constant velocity at the Tool Center Point (TCP) to ensure uniform material thickness. Any variation in speed at trajectory nodes results in non-uniform material accumulation.

a) Core Principle:

Dynamic path synthesis involves blending individual motion vectors into a continuous stream, ensuring the robot does not decelerate to a complete stop at intermediate waypoints.

b) Working Mechanism:

i) Arc Interpolation (Mvr): Unlike standard circular commands, Mvr generates smooth paths through three designated points (Start, Passed, and End), allowing for the construction of complex non-linear dispensing contours.

ii) Continuous Path Control (Cnt): By invoking the Cnt 1 command, the controller's algorithm blends the acceleration and deceleration phases of sequential commands. This ensures fluid motion transitions across via-points, preserving geometric integrity and maintaining constant feed rates.

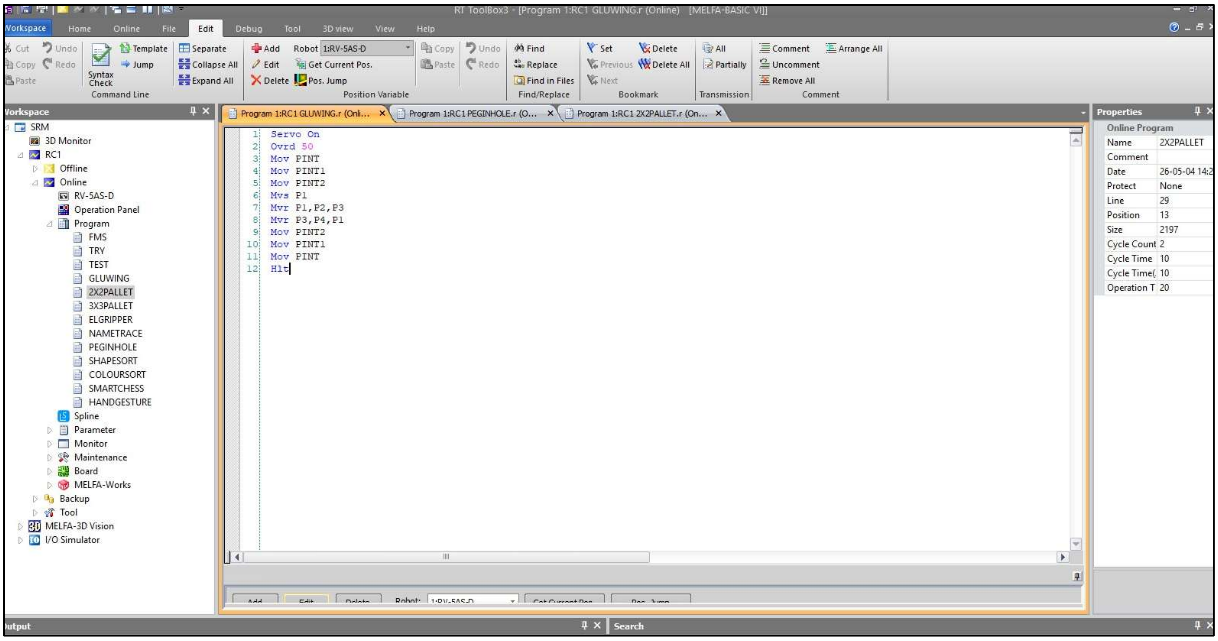

Figure 1: RT ToolBox3 editor showing MELFA-BASIC VI code with 3D circular interpolation (Mvr) for continuous dispensing.

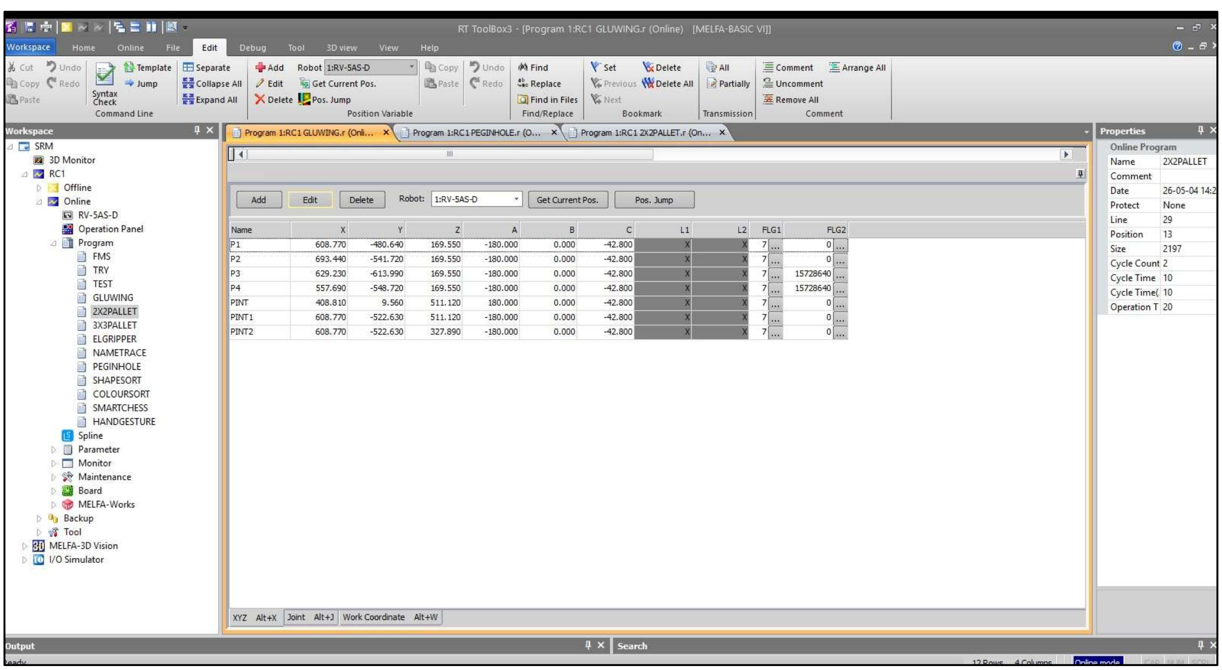

Figure 2: Position variable table with coordinates for the circular rim (P1–P4) and intermediate safety nodes (PINT series).

Figure 3: Workspace setup with Mitsubishi RV-5AS-D robot aligned over the circular fixture before dispensing.

Figure 3: End-effector performing linear approach to contact point (P1) before the 360° trajectory.

4. PROCEDURE:

a) Setup & Connections:

Establish communication between the workstation and the CR800-D controller. Interface the simulated dispenser trigger with the controller's digital output (e.g., Output 10). Define the absolute coordinates for the path entrance (p1), curve passing point (pPassed), and exit point (pEnd).

b) Algorithm & Logic:

i) Enable system servos and set the operational speed override (Ovrd) to a constant value suitable for dispensing.
ii) Enable continuous path blending using the Cnt 1 instruction.
iii) Navigate to a stabilized approach height above the start coordinate.
iv) Trigger the dispenser activation signal (M_Out).
v) Execute the combined linear (Mvs) and arc (Mvr) segments to synthesize the contour.
vi) Deactivate the dispenser signal at the terminal node and depart vertically to a safe origin.

c) Execution Steps:

i) Load the logic into RT ToolBox3 and perform a Step Run in manual mode to verify the approach/escape heights.
ii) Transition to Automatic mode and execute the full kinematic sequence.
iii) Open the Oscillograph tool and configure it to monitor Tool point speed (FB).

d) Observation Instructions:

Analyze the Oscillograph data to confirm that the feed rate remained stable during the transition from Mvs to Mvr. Observe the physical path for any mechanical vibration or "blobbing" at the intersection points.

5. RESULT AND OBSERVATION:

The robotic system successfully synthesized a continuous, high-fidelity dispensing trajectory. The implementation of Cnt 1 logic, combined with constant velocity override, ensured a stable feed rate across the arc segments without deceleration at intermediate nodes, as verified by Oscillograph feedback and visual inspection.

Expt No:
04
Date:
25 / 02 /2026

DISCRETE MATRIX PALLETIZING

1. AIM:

To synthesize a 4x4 spatial storage matrix using discrete grid logic within a simulated engineering environment, and to subsequently validate the dynamically calculated coordinate trajectories through live physical execution on a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

a) Robotic System: Mitsubishi RV-8CRL-D industrial manipulator.
b) Control Hardware: CR800-Series Controller and Teaching Pendant.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) Infrastructure: 16-point physical pallet base fixture mounted within the work envelope.
e) End-of-Arm Tooling (EOAT): Pointing effector or standard gripper configured to the tool coordinate system.

3. THEORY:

a) Core Principle:

In industrial automation, trajectory logic is first designed and simulated virtually to ensure mathematical accuracy and collision avoidance. Once the spatial discretization (the mathematical mapping of the physical workspace) is verified, the logic is deployed to the physical controller.

b) Working Mechanism:

i) Pallet Definition (Def Plt): The instruction establishes a localized 4x4 coordinate matrix mapped to Pallet 2. It is geometrically defined by physically teaching four absolute boundary coordinates (p14, p13, p12, p11) using the teaching pendant and syncing them to the software.

ii) Routing Pattern Generation: The final parameter in the Def Plt syntax (1) dictates the routing pattern. This forces the controller's algorithm to alternate the directional sequencing for each row (a zig-zag pattern), optimizing physical cycle time by minimizing unnecessary joint travel.

iii) Dynamic Coordinate Calculation: An index counter (m1) tracks grid saturation. The intermediate target variable (p100) is dynamically calculated and overwritten during each physical loop iteration.

c) Key Formula:

The absolute spatial target coordinate for the current cycle is derived dynamically by the controller:

$$\boldsymbol{P_{100} = Plt(2, m_1)}$$

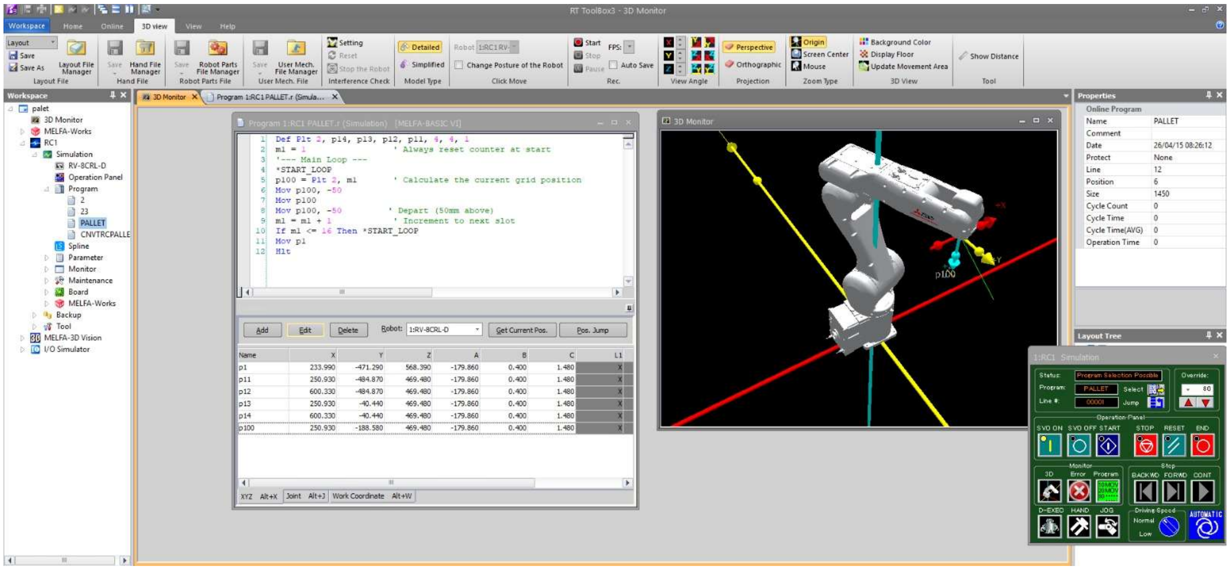

Figure 1: Integrated environment for programming, coordination, and simulation.

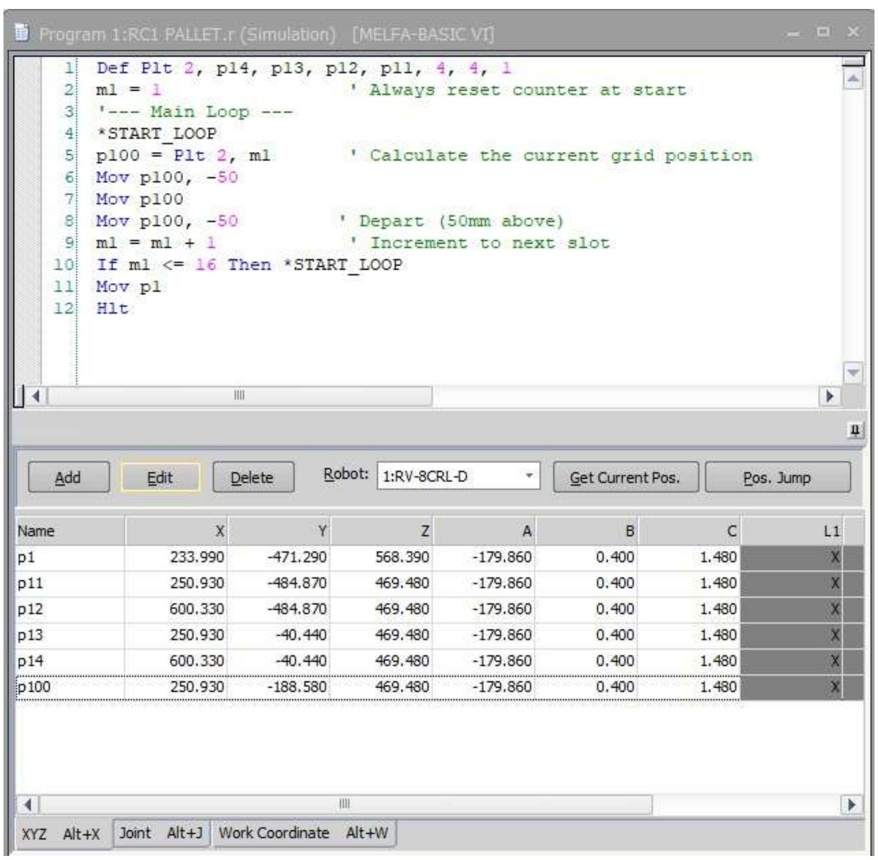

Figure 2: Grid iteration with defined parameters and boundary inputs.

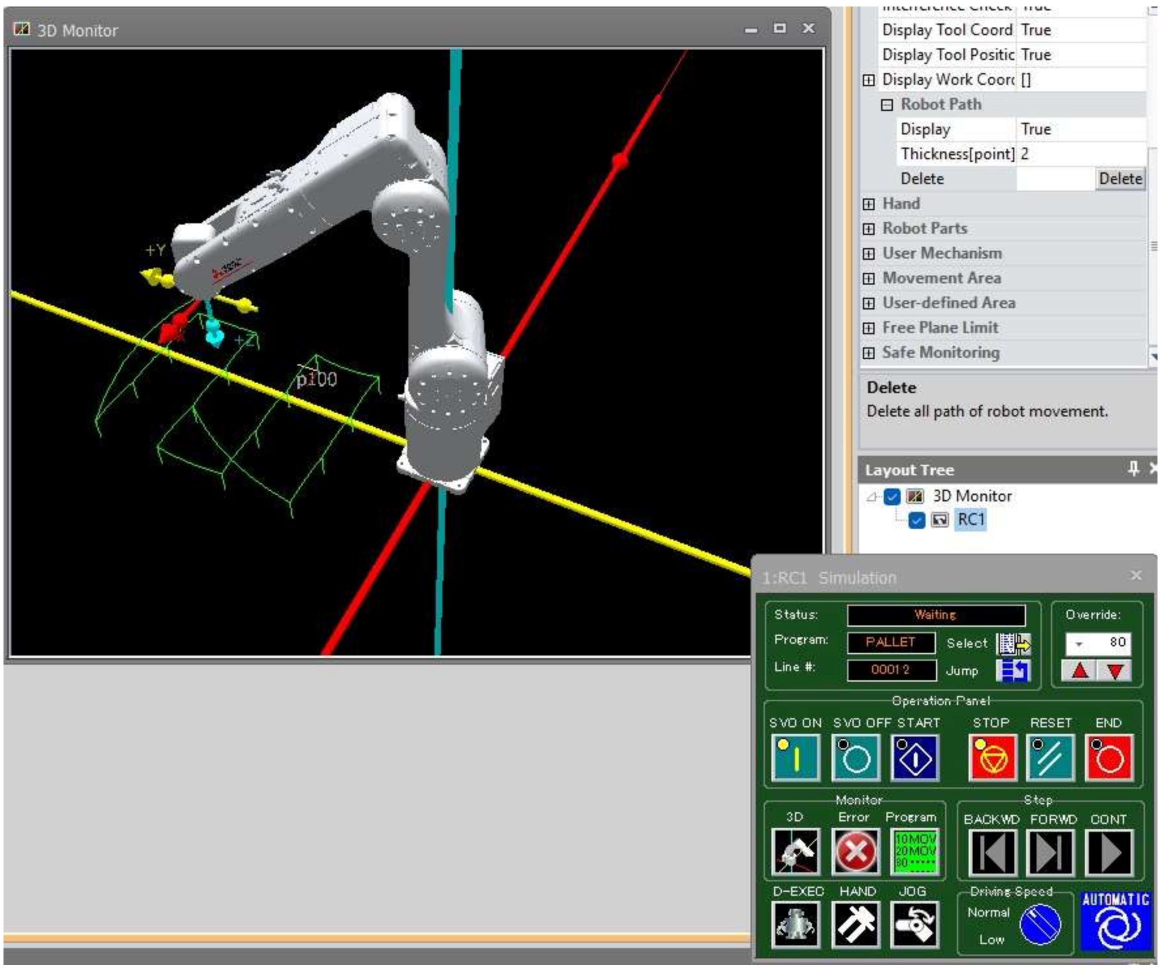

Figure 3: Simulated approach toward a dynamic target.

4. PROCEDURE:

a) Setup & Configuration:

Secure the physical pallet fixture. Using the teaching pendant, manually jog the RV-8CRL-D to teach the exact physical coordinates of the four corners (p11, p12, p13, p14) and a safe global origin (p1). Synchronize these coordinate registries with the RT ToolBox3 workspace.

b) Algorithm & Logic:

i) Declare the 4x4 pallet array (Pallet 2) with pattern type 1.
ii) Initialize the sequential counter (m1 = 1).
iii) Enter the main execution loop (*START_LOOP).
iv) Calculate the spatial placement node p100 for the current index.
v) Execute a joint-interpolated approach to p100 with a -50mm vertical Z-offset.
vi) Move linearly to the exact physical contact node p100.
vii) Execute a vertical linear departure to the -50mm Z-offset to clear the grid dividers.
viii) Increment the counter m1 and loop until the 16th node is traced.
ix) Return to the safe origin (p1) and halt execution.

c) Execution Steps:

i) Simulation Phase: Input the logic into the RT ToolBox3 Program Editor. Utilize the 3D Monitor to verify the toolpaths and ensure the -50mm offsets execute correctly without simulated collisions.
ii) Deployment: Write the verified program and position data directly to the physical CR800-D controller via the TCP/IP or USB interface.
iii) Physical Verification: Using the teaching pendant, initiate a Step Run at a low override speed (e.g., 10%) to visually verify the physical TCP aligns with the physical pallet fixture.
iv) Auto Execution: Transition the controller to Automatic mode and execute the full kinematic sequence at operational speed.

d) Observation Instructions:

During live execution, observe the mechanical smoothness of the zig-zag trajectory transitions. Verify that the physical end-effector perfectly aligns with the center of each of the 16 pallet nodes without any cumulative drift.

5. RESULT AND OBSERVATION:

The robotic system successfully validated the simulated 4x4 matrix palletizing logic through live physical execution. Dynamic coordinate mapping accurately guided the manipulator across the grid, and strict vertical approach/escape offsets ensured collision-free navigation, as verified by faculty evaluation.

Expt No:
05
Date:
05 / 03 /2026

DETERMINISTIC CALIBRATION AND ENCODER SYNCHRONIZATION

1. AIM:

To perform high-precision coordinate calibration between a linear transport system and a 6-axis robotic manipulator by mathematically mapping rotational encoder pulses to linear spatial displacement (P_EncDlt) on the Mitsubishi RV-8CRL-D industrial system.

2. APPARATUS REQUIRED:

a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
b) Control Hardware: CR800-D series dedicated robot controller.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) End-of-Arm Tooling : Precision drawing effector configured within the tool coordinate system.
e) Infrastructure: Solid installation platform and standard TCP/IP or standard USB communication interface.

3. THEORY:

Trajectory generation for planar symbolic mapping requires the mathematical decomposition of characters into primitive geometric segments.

a) Core Principle:

For an industrial robot to accurately interact with moving objects on a conveyor (Dynamic Conveyor Tracking), the controller must mathematically correlate the physical displacement of the conveyor belt with the Cartesian coordinate system of the manipulator. This requires establishing a precise pulse-to-millimeter ratio.

b) Working Mechanism:

i) Pulse Accumulation: As the conveyor moves, a connected rotary encoder generates digital pulses. The controller continuously reads this accumulated value via the M_ENC system variable.
ii) Spatial Reference Mapping: By physically aligning the robot with a target on the conveyor at two distinct locations (upstream and downstream) and recording both the absolute robot coordinates (P_CURR) and the encoder pulse counts at those exact moments, a mathematical slope vector is derived.
iii) Overflow Correction: Incremental encoders have a maximum logical threshold before the pulse count resets or rolls over. The algorithm must account for boundary crossings (e.g., the $\pm80,000,000$ pulse threshold) to maintain tracking integrity.

c) Key Formula:

The linear spatial displacement per single encoder pulse (P_EncDlt) across any axis is calculated as:

$$P_{EncDlt} = \frac{P_2 - P_1}{EncDiff}$$

Figure 1: Conveyor system with adjustable guide rails and photoelectric sensors for position mapping and trigger synchronization.

Figure 1: Lab setup with Mitsubishi RV-8CRL-D robot, encoder-based conveyor tracking system, and CR800-D controller.

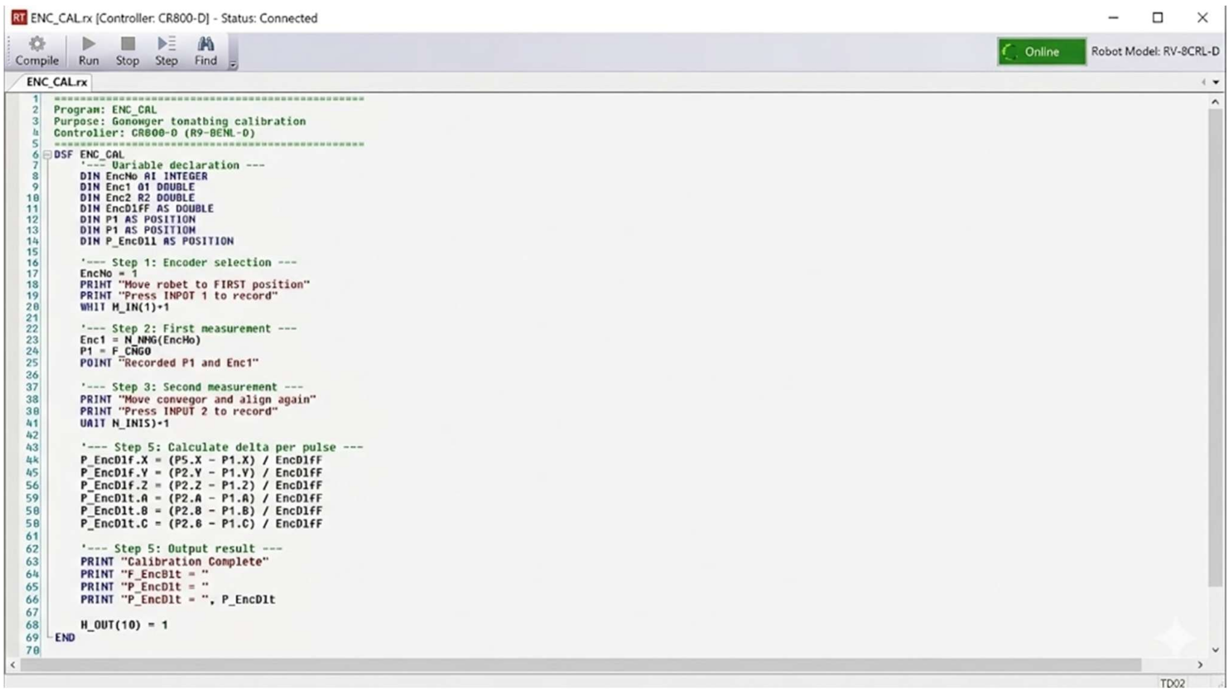

Figure : RT ToolBox3 Source Code for Conveyor Tracking Calibration.

4. PROCEDURE:

a) Setup & Connections:

Interface the conveyor's rotary encoder signal to the CR800-D controller's tracking card. Ensure the external trigger inputs (Input 1 and Input 2) are mapped to the teaching pendant or physical pushbuttons to capture the measurement snapshots.

b) Algorithm & Logic:

i) Initialize variables and select the target encoder channel (EncNo = 1).

ii) First Measurement: Jog the robot to a physical reference point on the upstream section of the conveyor. Trigger Input 1 to record the current position (P1) and pulse count (Enc1).

iii) Second Measurement: Run the conveyor to move the reference point downstream. Jog the robot to perfectly realign with the reference point. Trigger Input 2 to record the new position (P2) and pulse count (Enc2).

iv) Delta Calculation: Calculate the raw pulse difference (EncDiff) and apply the necessary overflow/underflow correction logic.

v) Compute the coordinate delta (P_EncDlt) for all 6 degrees of freedom (X, Y, Z, A, B, C).

c) Execution Steps:

i) Execute the ENC_CAL program in manual/step mode.

ii) Follow the generated PRINT prompts on the teaching pendant to align the robot TCP with the physical target and trigger the respective inputs.

iii) Verify the final calculated P_EncDlt matrix outputted on the pendant screen.

d) Observation Instructions:

Analyze the resulting coordinate delta values. Assuming the conveyor moves strictly along one primary Cartesian axis (e.g., the robot's Y-axis), the calculated delta for that specific axis should be a distinct non-zero value representing the millimeter-per-pulse ratio, while the remaining orthogonal axes (X, Z) and rotational postures (A, B, C) should theoretically calculate to zero.

5. RESULT AND OBSERVATION:

The system successfully derived the pulse-to-metric conversion factor P_{EncDlt}. The deterministic relationship established during calibration ensures that subsequent conveyor tracking operations maintain high spatial accuracy, as verified by functional evaluations.

Expt No:
06
Date:
12 / 03 /2026

AUTONOMOUS MATERIAL HANDLING AND MANIPULATION LOGIC

1. AIM:

To design and implement an automated pick-and-place robotic sequence using asynchronous sensor handshaking, effector actuation, and precise spatial trajectory mapping on a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

a) Robotic System: Mitsubishi RV-8CRL-D industrial manipulator.
b) Control Hardware: CR800-Series Controller.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) Sensing Hardware: Photoelectric workpiece detection sensor mapped to Input 6031.
e) Actuators: Component transport conveyor and pneumatic suction/gripper assembly actuated via Outputs 12/13.

3. THEORY:

A fundamental robotic pick-and-place operation requires the seamless integration of spatial motion planning with digital logic control to interact with the external environment.

a) Core Principle:

Autonomous manipulation relies on event-driven execution. The robot remains in a standby state until an external physical sensor triggers the initiation of a predefined sequence of kinematic sub-routines (retrieval, transfer, and placement).

b) Working Mechanism:

i) Approach/Escape Kinematics: To prevent lateral sheer collisions with fixtures or components during retrieval and placement, the kinematic sequence strictly enforces relative Z-axis offsets (approach and escape heights). This ensures the tool only enters the manipulation zone vertically.

ii) I/O Synchronization: The program architecture utilizes Wait M_In logic to halt motion execution until workpiece presence is verified by external instrumentation. Subsequent M_Out signals are triggered to electronically actuate pneumatic valves for payload securing and release.

iii) Point-to-Point Transfer: The sequence utilizes a combination of joint interpolation (Mov) for rapid spatial repositioning and linear interpolation (Mvs) for the precise vertical insertion into the pick and place coordinates.

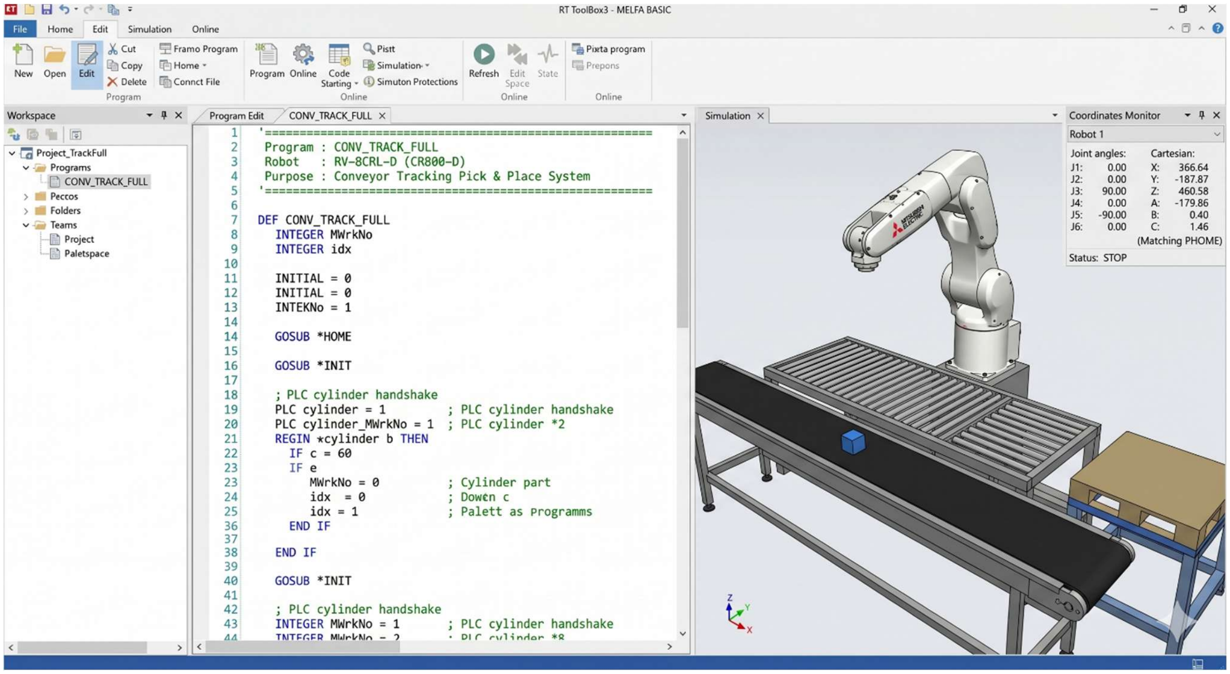

Figure 1: Program Initialization and Start of the 'CONV_TRACK_FULL' Loop.

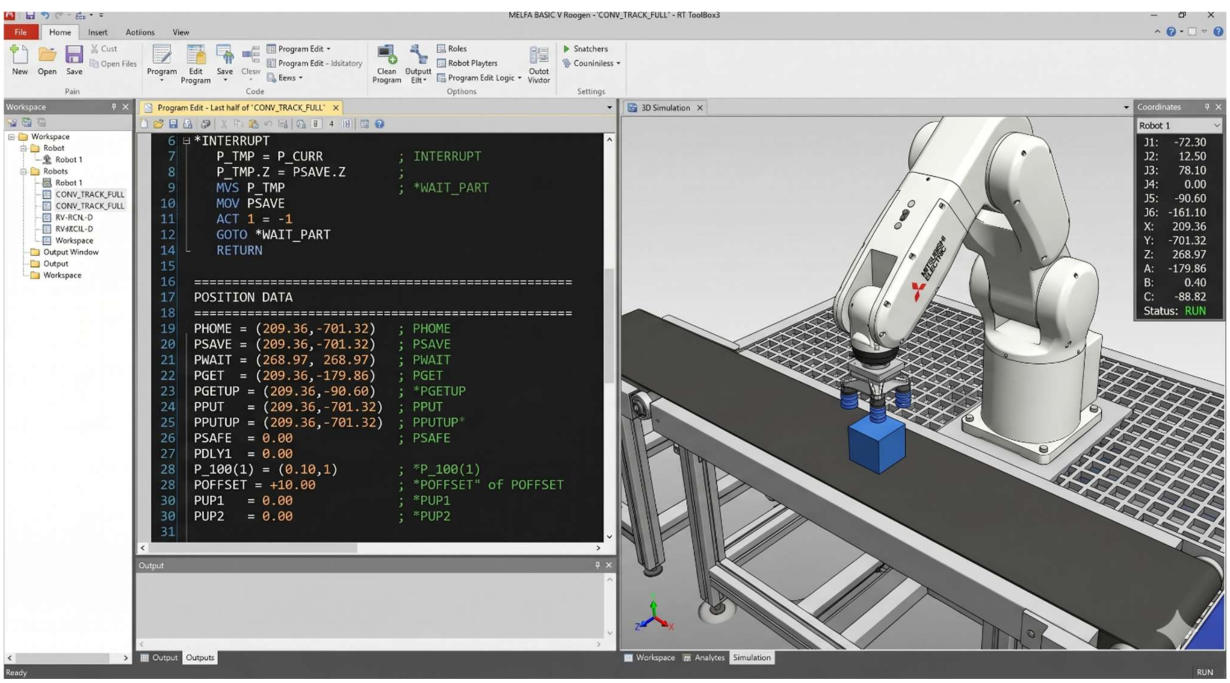

Figure 2: Complete Defined Position Coordinates and Variables Data Block.

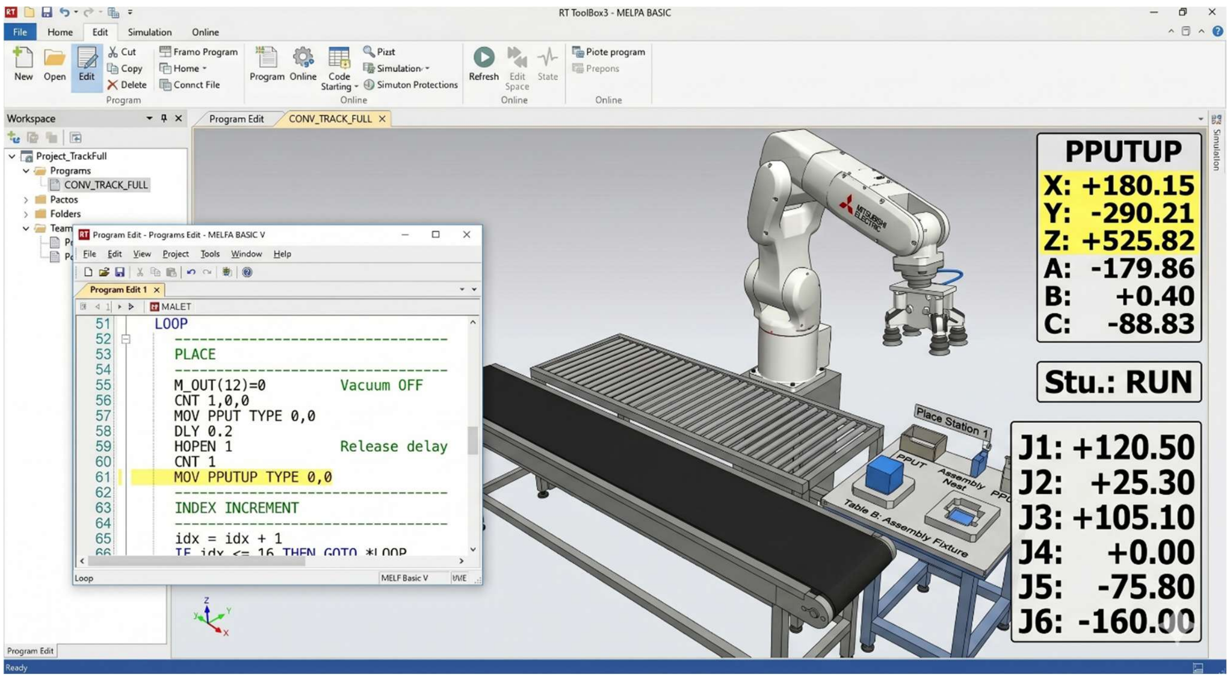

Figure 3: Maximized 3D Simulation Pane with Real-Time Coordinate Overlay at 'PGET' (Conveyor) Pick Station.

4. PROCEDURE:

a) Setup & Connections:

Interface the photoelectric sensor and pneumatic solenoid valves with the controller's I/O terminals. Configure the conveyor system and define the absolute spatial coordinates for the "Pick" (pPick) and "Place" (pPlace) stations.

b) Algorithm & Logic:

i) Activate the component conveyor and enter a continuous monitor loop for workpiece arrival.

ii) Upon photoelectric detection, halt the conveyor to stabilize the workpiece.

iii) Execute the kinematic "Pick" sequence: move to a safe Z-offset above the part, perform a linear approach, engage the pneumatic effector, and execute a vertical escape.

iv) Execute the "Place" sequence: move to a safe Z-offset above the drop zone, perform a linear insertion, disengage the pneumatic effector, and execute a vertical escape.

v) Reactivate the conveyor and loop the process for continuous automation.

c) Execution Steps:

i) Teach and record the absolute coordinates for pPick and pPlace using the teaching pendant.

ii) Verify the "Hand Type" parameter in RT ToolBox3 matches the installed pneumatic assembly logic.

iii) Validate the digital I/O handshake mechanism utilizing the Signal Monitor screen.

iv) Execute a Step Run to ensure the Z-offsets safely clear all environmental obstacles before switching to Automatic mode for full-speed execution.

d) Observation Instructions:

Analyze the operational Tact Time (cycle time from pick to place) and monitor the pneumatic grip stability during rapid joint interpolation (Mov) between the pick and place stations.

5. RESULT AND OBSERVATION:

The robotic system successfully performed synchronized pick-and-place operations with external conveyor and sensor systems. Proper approach/escape height control prevented interference, and reliable sensor handshaking ensured accurate part handling, as verified by faculty.

Expt No:
07
Date:
07 / 04 /2026

DUAL-GRID SYNCHRONIZATION AND SEQUENTIAL ROUTING LOGIC

1. **AIM:**
   To design and implement an automated inter-array transfer sequence that retrieves workpieces from a saturated 3x3 source matrix and systematically populates an empty 3x3 destination matrix utilizing a deterministic routing algorithm and intermediate safety nodes.

2. **APPARATUS REQUIRED:**
   a) Robotic System: Mitsubishi RV-8CRL-D industrial manipulator.
   b) Control Hardware: CR800-Series Controller and Teaching Pendant.
   c) Software Suite: RT ToolBox3 integrated engineering environment.
   d) Infrastructure: Precision-machined workspace table featuring two distinct 9-point (3x3) pallet grid fixtures (Source Grid and Destination Grid).
   e) End-of-Arm Tooling (EOAT): Two-finger parallel pneumatic gripper actuated via Output 15.

3. **THEORY:**
   a) Core Principle:
      Inter-array manipulation requires the simultaneous management of multiple localized coordinate systems. The manipulator must dynamically calculate both a retrieval node and a placement node during each operational cycle, ensuring geometric synchronization between the two physical grids while avoiding lateral collisions.
   b) Working Mechanism:
      i) Dual Pallet Definition: The system utilizes two independent Def Plt commands to map the Source (Pallet 1, using points P1-P4) and Destination (Pallet 2, using points P5-P8) matrices.
      ii) Routing Algorithm (Pattern 2): By setting the final parameter in the Def Plt syntax to 2, the controller's internal algorithm is forced to execute a specific non-standard deterministic sequence (e.g., zig-zag or columnar priority) across the 3x3 array, optimizing joint travel between indexing cycles.
      iii) Intermediate Trajectory Nodes (PInt): To safely navigate the physical space between the two separated grid plates without shearing the workpieces, the trajectory logic utilizes fixed intermediate waypoints (PInt, pInt1, PInt2). The robot must pass through these safe vertical clearance zones during every transfer.
   c) Key Formula:
      During each cycle index (M_1), the system calculates two distinct spatial coordinates simultaneously:

Pick = (Plt 1, M1)
Place = (Plt 2, M1)

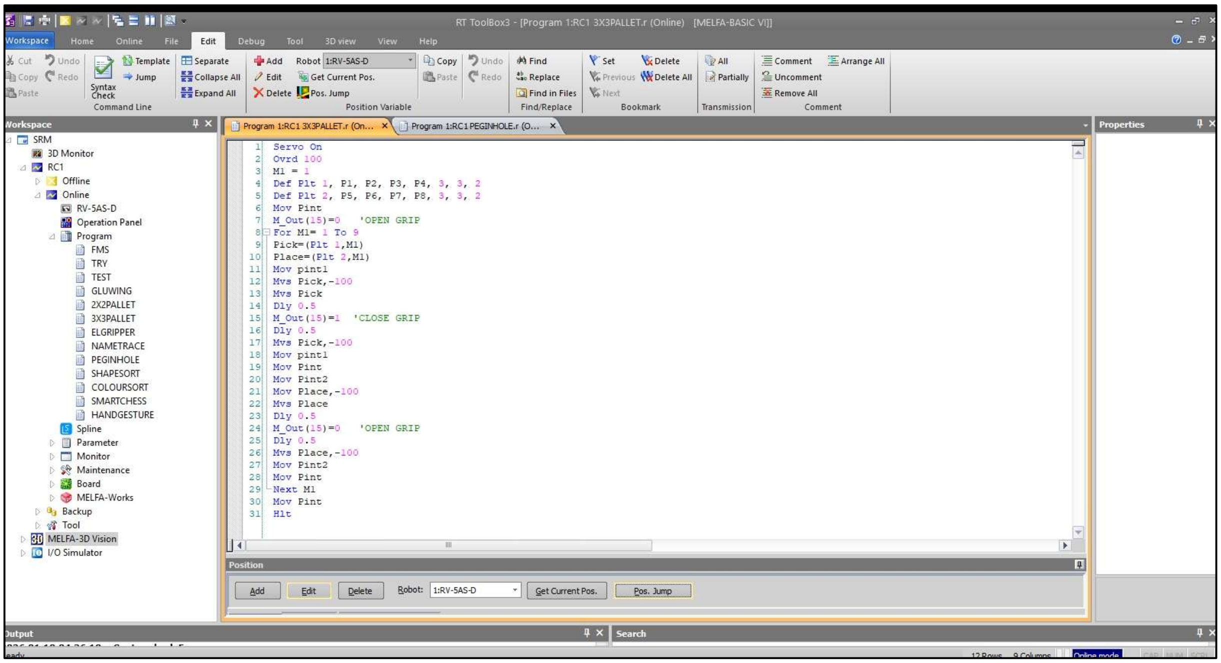

Figure 1: Program editor showing palletizing operation sequence.

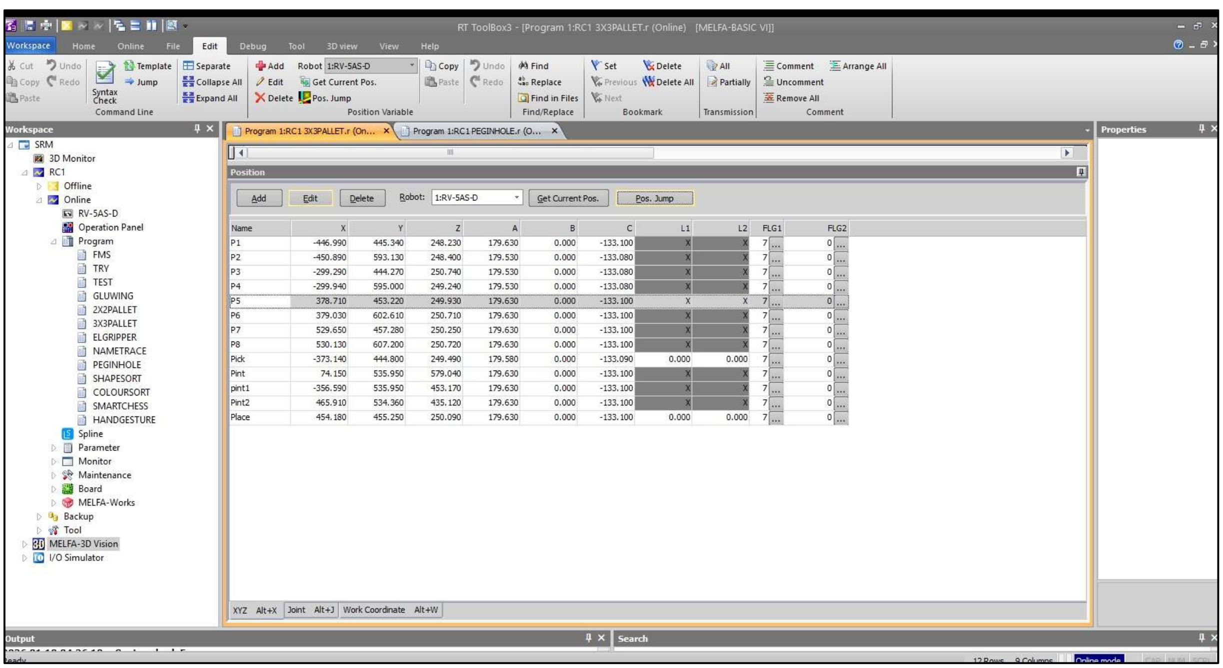

Figure 2: Position variables defining source and destination grid layout.

Figure 3: Manipulator performing workpiece pick from source grid.

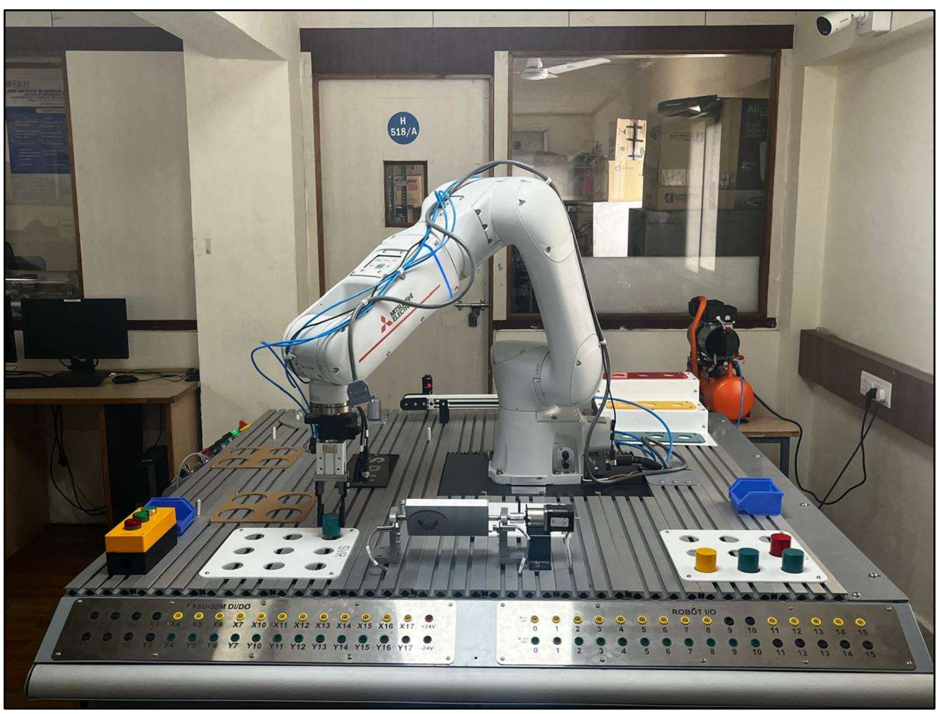

Figure 4: Manipulator performing workpiece placement in destination grid.

4. PROCEDURE:

a) Setup & Connections:

Ensure pneumatic pressure is supplied to the parallel gripper. Secure the red cylindrical workpieces in the 3x3 source grid. Using the teaching pendant, define the absolute boundary coordinates for Pallet 1 (P1 to P4) and Pallet 2 (P5 to P8). Critically, teach the safe intermediate clearance waypoints (PInt, pInt1, PInt2) high above the workspace.

b) Algorithm & Logic:

i) Initialize both 3x3 arrays using Def Plt with pattern type 2.

ii) Move to the global safe intermediate point (PInt) and ensure the gripper is open (M_Out(15)=0).

iii) Initiate a For...Next loop to index from 1 to 9.

iv) Retrieve: Calculate the Pick and Place coordinates. Move to the source-side intermediate point (pInt1), execute a vertical approach (-100mm), close the gripper to secure the workpiece, and retract vertically.

v) Transfer: Navigate back through pInt1 and the master safe point PInt to the destination-side intermediate point (PInt2).

vi) Place: Execute a vertical approach (-100mm) to the Place coordinate, open the gripper, and retract vertically.

vii) Iterate: Loop the process via Next M1 until all 9 objects are transferred, then return to PInt and halt.

c) Execution Steps:

i) Validate the physical coordinate registries for both pallets and all intermediate points in RT ToolBox3.

ii) Execute a Step Run in manual mode to verify that the spatial translation between pInt1, PInt, and PInt2 maintains sufficient Z-axis clearance over the grid hardware.

iii) Set Override speed to 100% (Ovrd 100) as defined in the program, and initiate Automatic mode execution.

d) Observation Instructions:

Visually monitor the kinematic sequence to confirm that the Pattern 2 parameter dictates the expected routing path across the grids. Observe the strict adherence to the intermediate routing nodes (PInt), ensuring no lateral sheer forces occur during the inter-array transfer.

5. RESULT AND OBSERVATION:

The system successfully executed the automated inter-array transfer between two distinct 3x3 storage grids. The implementation of the zig-zag routing algorithm optimized joint travel time, and the synchronized dual-coordinate calculations maintained strict positional accuracy throughout the sequence, as verified by faculty.

Expt No:
08
Date:
15 / 04 /2026

PROCESS ORCHESTRATION FOR MULTI-STAGE MANIPULATION

1. AIM:

To design and implement an integrated industrial workflow that orchestrates high-precision transfer sequences of flat circular workpieces between two distinct 2x2 matrices, utilizing deterministic state logic, vacuum-assisted End-of-Arm Tooling (EOAT), and strategically mapped clearance vectors.

2. APPARATUS REQUIRED:

a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
b) Control Hardware: CR800-D series dedicated robot controller.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) End-of-Arm Tooling : Vacuum suction cup end-effector actuated via a pneumatic generator on Output 14.
e) Infrastructure: Two precision-cut 2x2 wooden template fixtures (Source and Destination arrays) and four flat, circular workpieces (discs).

3. THEORY:

a) Core Principle:

Manipulating flat, flush-seated objects requires specialized vacuum-assisted tooling rather than mechanical grippers. Orchestrating this transfer between partitioned matrices demands the strict synchronization of continuous spatial kinematics (trajectory control) with discrete binary state logic (vacuum generation and purging).

b) Working Mechanism:

i) Deterministic State Logic for Vacuum Tooling: The actuation of the vacuum cup is governed by Boolean I/O mapping (M_Out(14)). Critical to this operation is the mathematical injection of temporal settling phases (Dly). Pneumatic vacuum generators experience inherent mechanical latency; delays must be enforced to allow the vacuum pressure to reach the theoretical gripping threshold (to establish a seal on the disc) before any Z-axis spatial displacement occurs.

ii) Dual-Array Kinematic Orchestration: The workflow abstracts the physical wooden templates into continuous mathematical algorithms using two Def Plt commands. By isolating the source and destination grids with master intermediate routing nodes (Pint, pint1, Pint2), the sequence enforces strict vertical entry and departure vectors (e.g., Mov Pick, -100). This completely eliminates lateral sheer forces, preventing the discs from catching on the edges of the wooden templates during extraction and insertion.

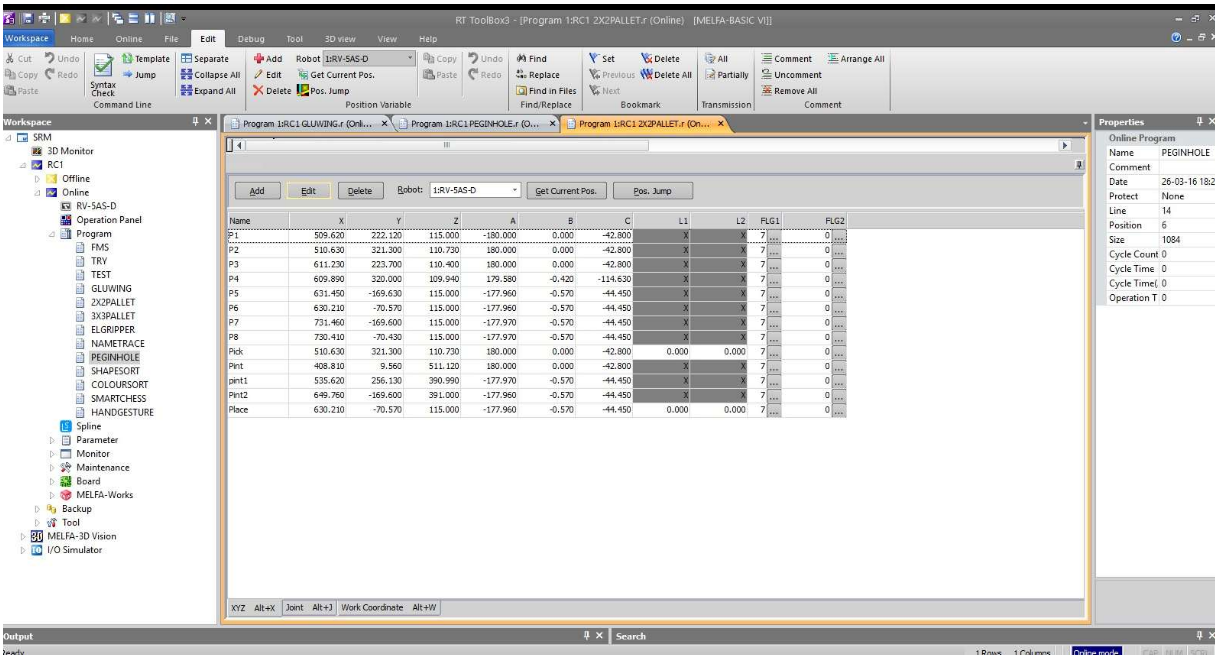

Figure 1: RT ToolBox3 position table defining 2×2 matrices (P1–P8) and intermediate routing points.

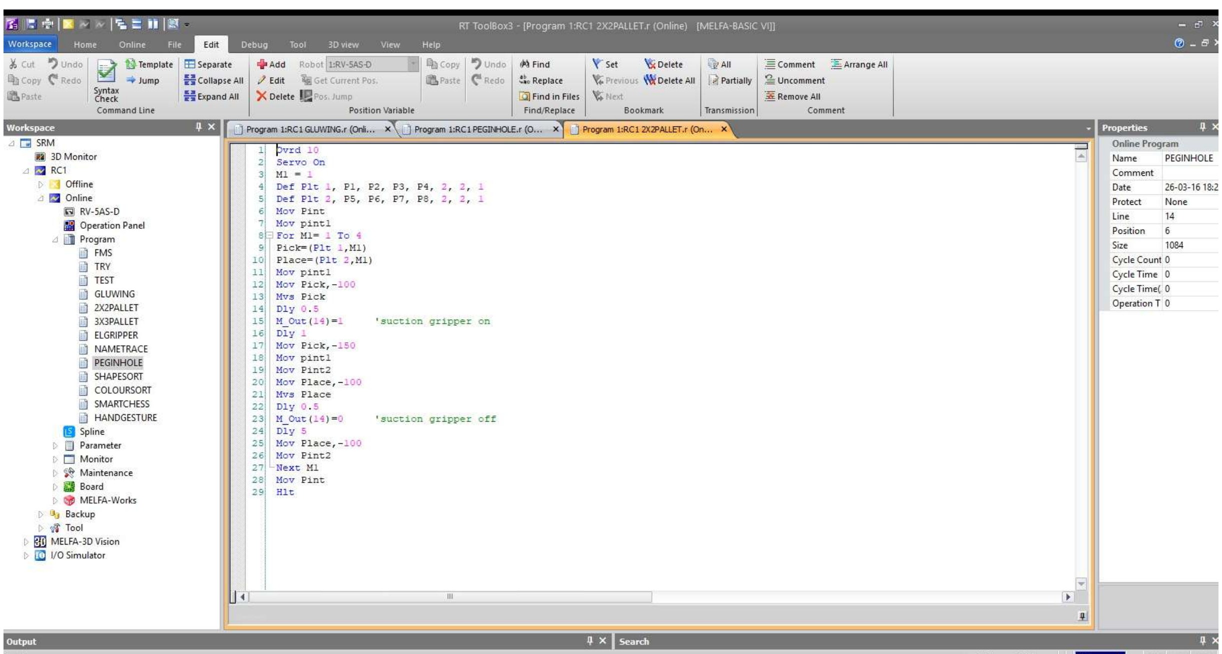

Figure 2: MELFA-BASIC VI execution sequence showing control logic and timing delays.

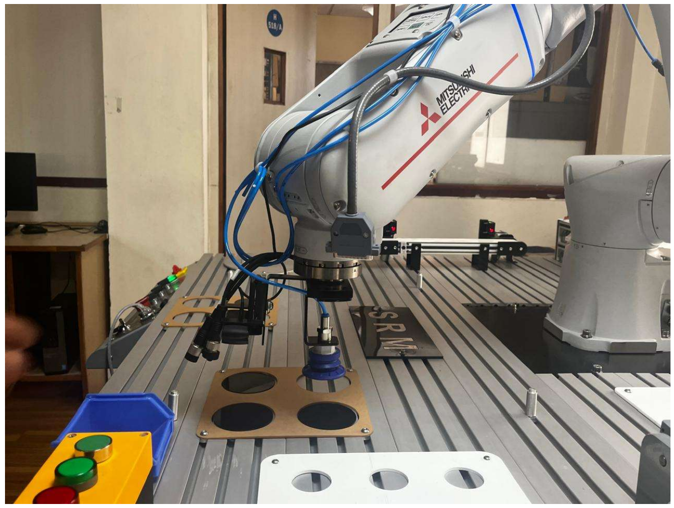

Figure 3: Gripper retrieving a circular workpiece from the source matrix.

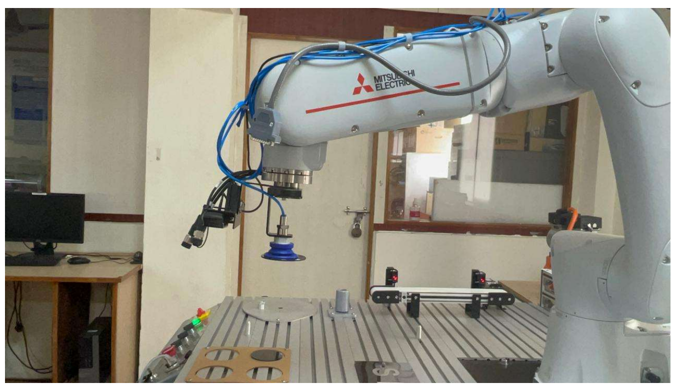

Figure 3: Manipulator transferring the workpiece through a safe intermediate path.

Figure 3: Vertical placement of the workpiece into the destination matrix.

4. PROCEDURE:

a) Setup & Connections:

Interface the vacuum generator to Output 14. Seat the four circular discs into the source wooden template. Using the teaching pendant, define the exact surface-contact center points for the 2x2 source matrix (P1 to P4) and the destination matrix (P5 to P8). Establish the intermediate safety nodes (Pint, pint1, Pint2) well above the fixture profiles.

b) Algorithm & Logic:

i) Throttle the global velocity override to 10% (Ovrd 10) to prioritize spatial precision during the vacuum seal phase.
ii) Define the geometry for both 2x2 matrices utilizing pattern type 1.
iii) Initiate a localized loop iterating from index 1 to 4 (For M1= 1 To 4).
iv) Retrieval State: Compute localized Pick and Place coordinates. Route the TCP through the source-side intermediate node (pint1). Execute a linear plunge to the disc surface.
v) Transition the state logic (M_Out(14)=1) to generate vacuum pressure. Dwell for 1.0 second to establish the seal, then retract vertically by 150mm.
vi) Transfer State: Route the secured payload through the high-clearance intermediate nodes (pint1 → Pint2).
vii) Placement State: Execute a vertical plunge (-100mm) into the destination template. Transition the state logic (M_Out(14)=0) to purge the vacuum. Dwell for 5.0 seconds to guarantee complete pressure equalization and payload release, then retract vertically.
viii) Iterate the loop until total matrix transfer is achieved.

c) Execution Steps:

i) Compile the sequence and ensure the vacuum lines are pressurized and free of mechanical kinks.
ii) Perform a Step Run to verify the temporal delays (Dly 1 and Dly 5) provide adequate pneumatic latency compensation for the specific mass of the black discs.
iii) Execute the sequence autonomously, ensuring the vertical offsets completely clear the wooden fixture templates during transfer.

d) Observation Instructions:

Analyze the kinematic interplay between the state transition of Output 14 and the immediate subsequent Z-axis vertical lift. Observe the integrity of the vacuum seal during the rapid joint-interpolated transfer across the workspace.

5. RESULT AND OBSERVATION:

The system successfully transferred circular workpieces between two 2×2 arrays. Deterministic logic and timed delays enabled reliable vacuum gripping, while strict vertical motion prevented contact with the template boundaries, as verified by faculty.